



Gen AI Academy

APAC Edition

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Participant Details

Participant Name: Inclusive Intelligence Lab

Problem Statement: AccessPrep AI - AI-powered Decision Intelligence for Accessible Competitive Exam Preparation for visually impaired students.

Brief about the idea

AccessPrep AI converts inaccessible images and PDFs into OCR text, Gemini explanations, follow-up tutoring, MCQs, audio output, and personalized study plans for TNPSC, Railway, SSC, Banking, and UPSC aspirants. It is voice-first, screen-reader friendly, and Cloud Run ready.

- Approach: We translated the accessibility problem into a working Google Cloud prototype using Streamlit, Google Vision OCR or Gemini Vision, Gemini tutoring, optional Vertex AI Gemini, BigQuery learning analytics, and Cloud Run deployment.
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Real-world impact: Visually impaired learners gain independent access to study material, explanations, quizzes, and revision planning.

Workflow: Upload -> OCR -> concept explanation -> chat Q&A -> MCQ generation -> decision intelligence -> speech output.

Approach, impact, and intelligent workflow

Opportunities and USP

- USP: AccessPrep AI is not only an OCR tool. It is an accessible exam-preparation decision engine that combines content extraction, conversational AI, quiz generation, study prioritization, confidence scoring, and voice output.
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- Differentiation: Built specifically for visually impaired competitive exam aspirants with large fonts, dark mode, speech-friendly responses, and personalized study recommendations.

List of features offered by the solution

1. Image and PDF upload with preview
2. OCR extraction using Google Vision OCR or Gemini Vision
3. Gemini concept explanation and summarization
4. Follow-up chat interface
5. Five MCQ generation with answers
6. Weak-topic and next-chapter recommendation
7. Seven-day revision and study plan
8. Confidence score
9. gTTS speech output
10. Cloud Run ready deployment

Process flow diagram or use-case diagram

Student uploads image/PDF -> Streamlit validates and previews file -> OCR layer extracts text -> Gemini tutor explains, summarizes, chats, and creates MCQs -> Decision engine combines exam, days remaining, study hours, score, and extracted text -> Personalized recommendations are shown -> gTTS converts responses to speech -> optional BigQuery event logging captures anonymized learning events.

Wireframes / mock diagrams of the proposed solution

Implemented screens:

Home: project, problem, solution, team, hackathon context

Upload: image/PDF upload, preview, OCR extraction, extracted text

AI Tutor: explain, summarize, key points, MCQs, chat, speech

Study Recommendation: profile inputs, weak topics, next chapter, schedule, confidence

About: architecture, technology, future scope, GitHub readiness

Architecture diagram of the proposed solution

The repository includes architecture.png with the full Google Cloud style architecture. Components: Student interface, Streamlit UI, OCR layer, Gemini tutor, decision engine, accessibility speech layer, BigQuery analytics, Cloud Run container deployment, Artifact Registry, Secret Manager, and Cloud Logging.

Technologies / Google / NVIDIA services used in the solution

Core stack: Python, Streamlit, Gemini API, Google Vision OCR, Gemini Vision, python-dotenv, gTTS, Pillow, PyMuPDF, Docker.

Google Cloud stack: Cloud Run for hosting, Vertex AI Gemini readiness for managed inference, Vision API for OCR, BigQuery for anonymized learning analytics, Artifact Registry for images, Secret Manager for API keys, Cloud Logging for observability, Cloud Build for CI/CD.

Why this stack: scalable serverless deployment, managed AI services, low operational overhead, strong accessibility workflow, and production-oriented observability. NVIDIA services are not used because this solution targets the Google Cloud Gen AI track.

Prototype implemented in the repository:

- Snapshots of the prototype
Home page with project and team details
- **Upload page with image/PDF preview and OCR result**
- **AI Tutor page with explain, summarize, MCQ, and chat actions**
- **Study Recommendation page with student profile and confidence output**
- **About page with architecture and future scope**

Run with: streamlit run app.py

Google Cloud 

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Thank you

